Bharathraaj Nagarajan

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Work Experience

University at Buffalo – AI Innovation Lab (A2IL) | *Volunteer Research Scientist* Apr 2026 – Present

- Built and maintained backend services for a faculty activity intelligence platform using **Python and FastAPI**, exposing 9 REST API endpoints over a 12-table **MySQL** schema with an asynchronous data-access layer, integrating ∼11 external data sources to track 65 faculty profiles.
- Contributed to a stateful **LangGraph** workflow orchestrating multi-provider **LLM** routing to produce structured JSON outputs from the ingested multi-source data.
- Containerized and deployed backend and frontend services on **Docker and OpenShift/Kubernetes**, configuring Deployments, Services, Routes, ConfigMaps, and Secrets with readiness/liveness health checks; diagnosed and resolved container permission, path-resolution, and connectivity/timeout failures to stabilize production rollout.

Media Sales Plus Inc., Buffalo, NY | *Software Engineer Intern – Backend & Cloud* Aug 2025 – Dec 2025

- Built backend data ingestion and transformation pipelines in **Python**, integrating 10+ multi-tenant client datasets across heterogeneous schemas on **Azure and GCP**, exposing the processed data through backend services for downstream analytics.
- Engineered **CI/CD pipelines** using GitHub Actions and Docker to automate deployment of backend services, achieving zero-downtime releases and eliminating manual release overhead.
- Extended the backend platform with an **LLM-powered RAG query service** over structured business data, integrating vector retrieval and inference calls into existing APIs for real-time stakeholder queries.

Lloyds Banking Group | *Software Engineer – Distributed Systems* Oct 2021 – Apr 2024

- Designed and optimized distributed backend pipelines using **Spark (PySpark, Spark SQL), Kafka, HDFS, HBase, and Kudu** across banking systems, processing **100M+ records/day** and reducing query latency by ∼**40%** through partitioning strategies and query optimization.
- Engineered high-availability cloud-native backend services on **Kubernetes**, achieving **99.9% uptime** via containerized deployment, load balancing, automated recovery, and end-to-end monitoring.
- Designed scalable, fault-tolerant microservice architectures for distributed workflows, translating business requirements into reliable services and resolving production failures under strict SLA constraints.
- Optimized large-scale batch and streaming pipelines, improving processing efficiency by ∼47% through parallelization, partition tuning, and performance profiling across distributed nodes.

Projects

Durable & Governed Agent Execution | *Python, Temporal, Claude, MongoDB, Keycard, FastAPI | Link*

- Built a **durable AI agent execution system** integrating Claude structured outputs, deterministic policy, Temporal workflows, human approval, and verified remediation actions with retrieval over **MongoDB Vector Search**.
- Implemented **per-attempt Keycard credentials, idempotent execution, retries, and audit records**; validated reliability across **12/12 live failure scenarios** with 162 unit/integration tests and 5 live Keycard tests.

LLM Inference & GPU Serving Platform | *Python, PyTorch, vLLM, CUDA, Docker, Kubernetes | Link*

- Built a **GPU-accelerated LLM inference service** using PyTorch and vLLM on NVIDIA CUDA GPUs, exposing an **OpenAI-compatible API** and containerizing the serving stack with Docker for deployment on Kubernetes.
- Implemented **concurrent inference benchmarking** across batch sizes and request loads, measuring **latency, throughput, and GPU utilization** to evaluate serving-performance tradeoffs and characterize GPU inference efficiency.

AI-Powered Patient Scheduling Portal | *Next.js, TypeScript, GPT-4o, AWS EC2, Nginx | Link*

- Built and deployed a **full-stack AI application** integrating GPT-4o-powered patient intake and semantic doctor matching with deterministic workflow/state management and **REST backend services**.
- Deployed on **AWS EC2** with Nginx reverse proxy and PM2 process management, integrating production hosting and process management for the full-stack application.

Technical Skills

Programming Languages: Python, Java, C++, C, SQL, TypeScript
Backend Engineering: FastAPI, REST APIs, API Design, Microservices, Service Architecture, Asynchronous Programming, Object-Oriented Programming
Distributed Systems & Data: Apache Spark (PySpark, Spark SQL), Kafka, Hadoop, HDFS, HBase, Kudu, Batch Processing, Stream Processing, Partitioning, Parallel Processing, Fault Tolerance, Scalability
AI / Agentic Systems: LLM APIs, LLM Integration, RAG, Vector Retrieval, LangGraph, Agentic Workflows, Structured Outputs, Tool Execution / MCP
ML / AI Infrastructure: PyTorch, vLLM, CUDA, RAPIDS cuDF, MLflow, LLM Inference, GPU-Accelerated Data Processing
Cloud & Platform: Kubernetes, OpenShift, Docker, Containerization, AWS (EC2, S3), Azure, Google Cloud Platform, Nginx, GitHub Actions, CI/CD, Linux
Production & Reliability: High Availability, Monitoring, Observability, Health Checks, Retries, Timeout Handling, Load Balancing, Performance Profiling, Performance Optimization, Production Debugging, Incident Response, SLAs
Databases: MySQL, PostgreSQL, Snowflake, BigQuery, Azure SQL

Education

University at Buffalo, The State University of New York Aug 2024 – Dec 2025

Masters in Data Sciences and Applications GPA: 3.9/4.0 *Buffalo, NY*

- Coursework: Data Intensive Computing, Distributed Systems, Machine Learning, Database Management, Cybersecurity

SSN College of Engineering Aug 2017 – May 2021

Bachelor of Engineering, Minor in Computer Science *Chennai, India*